

Task 1:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
GNU nano 7.2 producer_consumer.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

int main() {
    int pipefd[2];
    int data;
    pid_t pid;

    pipe(pipefd);

    pid = fork();

    if (pid > 0) {
        // Parent = Producer
        close(pipefd[0]);

        clock_t start = clock();

        for (int i = 1; i <= 5; i++) {
            write(pipefd[1], &i, sizeof(i));
            printf("Produced: %d\n", i);
        }

        close(pipefd[1]);
        wait(NULL);

        clock_t end = clock();

        printf("Time taken: %f seconds\n",
              (double)(end - start) / CLOCKS_PER_SEC);
    } else {
        // Child = Consumer
        close(pipefd[1]);

        while (read(pipefd[0], &data, sizeof(data)) > 0) {
            printf("Consumed: %d\n", data);
        }

        close(pipefd[0]);
    }

    return 0;
}
```

Output:

```
samanth-ubuntu@samanth-ubuntu:~$ nano producer_consumer.c
samanth-ubuntu@samanth-ubuntu:~$ gcc producer_consumer.c -o producer_consumer
samanth-ubuntu@samanth-ubuntu:~$ ./producer_consumer
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Produced: 5
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Time taken: 0.000178 seconds
samanth-ubuntu@samanth-ubuntu:~$
```

Observation:

The parent process acts as the **producer** and generates integer values from 1 to 5. These values are sent to the child process through an **anonymous pipe** using the write() system call. The child process acts as the **consumer** and receives the values using the read() system call. The communication time is measured to determine the efficiency of data transfer between the two processes.

Result:

The producer-consumer communication system was successfully implemented using an **anonymous pipe**, and the communication time was measured successfully.

Task 2:

Develop a program that executes the equivalent of the shell command `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
GNU nano 7.2                                pipe_ls.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipefd[2];

    pipe(pipefd);

    if (fork() == 0) {
        // Child 1: ls -l
        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);
    }
    else {
        // Parent creates second child
        if (fork() == 0) {
            // Child 2: grep ".c"
            close(pipefd[1]);

            dup2(pipefd[0], STDIN_FILENO);
            close(pipefd[0]);

            execlp("grep", "grep", ".c", NULL);
        }

        close(pipefd[0]);
        close(pipefd[1]);

        wait(NULL);
        wait(NULL);
    }

    return 0;
}
```

Output:

```
samanth-ubuntu@samanth-ubuntu:~$ nano pipe_ls.c
samanth-ubuntu@samanth-ubuntu:~$ gcc pipe_ls.c -o pipe_ls
samanth-ubuntu@samanth-ubuntu:~$ ./pipe_ls
-rwxr-xr-x 1 samanth-ubuntu samanth-ubuntu 4096 Sep 28 09:40 escape
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 811 Sep 28 09:40 escape.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 946 Sep 28 09:40 fork_example.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 841 Sep 28 09:40 hello.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 4756 Sep 28 09:40 main.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 2755 Sep 28 09:40 myshell.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 694 Sep 28 09:40 pipe_ls.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 16312 Sep 28 09:40 process
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 810 Sep 28 09:40 process.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 16384 Sep 28 09:40 process_state
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 1248 Sep 28 09:40 process_state.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 16312 Sep 28 09:40 producer_consumer
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 839 Sep 28 09:40 producer_consumer.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 35 Sep 28 09:40 script.sh
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 338 Sep 28 09:40 state.c
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 1517 Sep 28 09:40 wait_compare
-rw-r--r-- 1 samanth-ubuntu samanth-ubuntu 784 Sep 28 09:40 zombie.c
samanth-ubuntu@samanth-ubuntu:~$
```

Observation:

The program successfully implements the shell pipeline using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls. The output of `ls -l` is redirected through the pipe to `grep ".c"`, which displays the matching entries.

Result:

The shell command `ls -l | grep ".c"` was successfully implemented using system calls.